

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2022

TEST DATE: 27-02-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. $I = \frac{E}{3R}$

In loop PSQP,

$$E - \frac{Q_1}{C_1} + E = 0$$

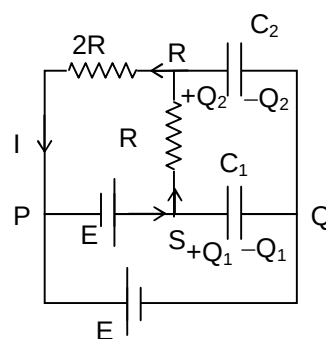
$$\Rightarrow Q_1 = 2EC_1$$

$$\text{Also, } E - IR - \frac{Q_2}{C_2} + E = 0$$

$$Q_2 = \frac{5EC_2}{3}$$

$$\text{Also, } \frac{Q_1^2}{2C_1} = \frac{Q_2^2}{2C_2}$$

$$\Rightarrow \frac{C_2}{C_1} = \left(\frac{Q_2}{Q_1} \right)^2 = \frac{25}{36}$$



2. C

Sol. Applying KVL, we can get

$$I_1 = \frac{34}{13} \text{ A and } I_2 = \frac{36}{13} \text{ A}$$

$$\therefore I_2 - I_1 = \frac{2}{13} \text{ A}$$

3. D

Sol. $P = P_1 + P_2$

$K = K_1 + K_2$

$$\frac{P^2}{2m} = \frac{P_1^2}{2m} + \frac{P_2^2}{2m}$$

Solving, we get

$$P_1 = -\frac{P}{3}, \quad P_2 = \frac{4P}{3}$$

$$k_1 = \frac{k}{9}, \quad k_2 = \frac{8k}{9}$$

4. B

Sol. Time period $T = 2\pi\sqrt{\frac{2m}{3k}}$

Required time is $t = \frac{T}{4}$

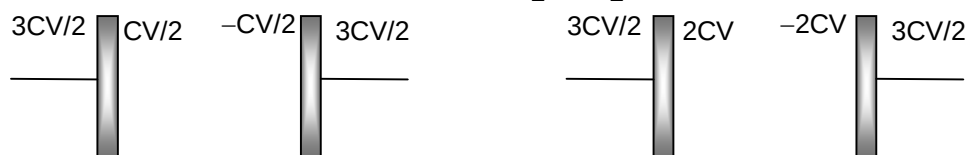
$$\therefore t = \frac{\pi}{2}\sqrt{\frac{2m}{3k}}$$

5. C

Sol. Phase difference is $\frac{\pi}{2} + \tan^{-1}\left(\frac{X_L}{R}\right)$

6. D

Sol. Hence charge flowing the battery is $2CV - \frac{CV}{2} = \frac{3CV}{2}$



7. A

Sol. In this case angular frequency is

$$\omega = \sqrt{\frac{16k}{9m}}$$

$$\therefore T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{9m}{16k}}$$

8. C

Sol. $\frac{1}{\lambda_0} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$

$$\Rightarrow n = \sqrt{\frac{R\lambda_0}{R\lambda_0 - 1}}$$

9. B

 Sol. At time t , velocity of 1st is

$$\vec{v}_1 = 30\hat{i} - gt\hat{k}$$

From 2nd, $\vec{v}_2 = 30\hat{j} - g t \hat{k}$

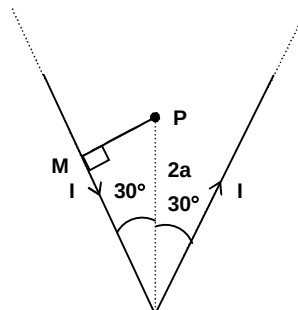
Also, $\vec{v}_1 \cdot \vec{v}_2 = v_1 v_2 \cos 60^\circ$

Solving we get $t = 3$ sec

10. C

Sol. $\vec{B} = \frac{2\mu_0 I}{4\pi a} (\sin 60^\circ + \sin 90^\circ) \hat{i}$

$$\vec{B} = \frac{\mu_0 I}{4\pi a} (2 + \sqrt{3}) \hat{i}$$



11. B

Sol. $dQ = dU + dW = 3dU$

$$dQ = 3nC_V dT = nCdT$$

$$\Rightarrow C = 3C_V$$

$$\Rightarrow 7.5 R = 3C_V$$

$$\therefore C_V = \frac{5R}{2}$$

Gas is diatomic

12. D

Sol. when $m = 256$ gm

$$120 = \frac{3}{2 \times 1.5} \sqrt{\frac{256g}{\mu}} \quad \dots(i)$$

When $m = m$, then

$$120 = \frac{4}{2 \times 1.5} \sqrt{\frac{mg}{\mu}} \quad \dots(ii)$$

Solving (i) and (ii), we get

$$m = 144 \text{ gm}$$

13. D

Sol. If ℓ = length of spring

$$\ell = \ell_A + \ell_B \quad \dots(i)$$

$$\ell_A : \ell_B = 2 : 3 \quad \dots(ii)$$

From (i) and (ii)

$$\ell_A = \frac{2\ell}{5}, \ell_B = \frac{3\ell}{5}$$

$$k \times \ell = kb \times \frac{3\ell}{5}$$

$$\Rightarrow kb = \frac{5k}{3}$$

14. C

Sol. It will be equal to $\sqrt{2gh}$

15. B

Sol. for rotation

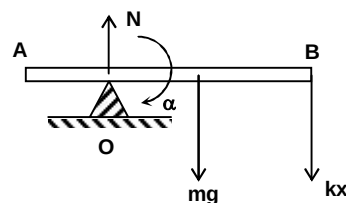
$$mg(0.1) + kx(0.6) = I\alpha$$

$$\text{Solving, } \alpha = 30 \text{ rad/s}^2$$

For translational

$$mg + kx - N = m(0.1)\alpha$$

$$\Rightarrow N = 20 \text{ newton}$$



16. A

Sol. Net force

$$\vec{F} = q(\vec{E}_1 + \vec{E}_2) + q\{\vec{v} \times (\vec{B}_1 + \vec{B}_2)\}$$

$$\therefore \vec{F} = qE_0(0.8\hat{i} + \hat{j} + 0.2\hat{k})$$

 Here at $t = 0$, $x = y = 0$

$$\vec{E}_1 = E_0\hat{i}, \quad \vec{E}_2 = E_0\hat{k}$$

$$\vec{B}_1 = \frac{E_0}{C}\hat{k}, \quad \vec{B}_2 = \frac{E_0}{C}\hat{i}$$

17. B

$$\text{Sol. } F = (1+r)\frac{IA}{C}$$

$$= \frac{(1+0.25)(25)(2)}{3 \times 10^8} \approx 20 \times 10^{-8} \text{ N}$$

18. D

 Sol. When potential energy is $\left(\frac{1}{4}\right)^{\text{th}}$ its maximum value, then current through 24 V battery is

$$I = \frac{21}{8} \text{ A}$$

$$\therefore \text{Power} = VI = 24 \times \frac{21}{8} \text{ W} = 63 \text{ W}$$

19. B

 Sol. If a beam of unpolarized light of intensity I_0 is passed through the two polaroids, the intensity of the emergent light is

$$I = \frac{I_0}{2} \cos^2 \theta$$

$$I = \frac{I_0}{2} \cos^2 30^\circ = \frac{3I_0}{8}$$

20. D

Sol. Truth table for given combination is

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

This comes out to be truth table of OR gate

SECTION – B

21. 00003.00

Sol. $I_1 = \frac{E}{R} e^{-t/RC}$

$$I_2 = \frac{E}{R} \left(1 - e^{-\frac{tR}{L}} \right)$$

$$\frac{I_2}{I_1} = \frac{1 - e^{-2t}}{e^{-2t}} = 3$$

22. 00019.00

Sol. $P_1 = \frac{4T}{a}, V_1 = \frac{4}{3}\pi a^3$

$$P_2 + \frac{\sigma^2}{2\epsilon_0} = \frac{4T}{2a}, V_2 = \frac{4}{3}\pi(2a)^3$$

$\therefore$ Process is adiabatic,

$$P_1 V_1^{5/3} = P_2 V_2^{5/3}$$

Solving we get

$$\sigma = \sqrt{\frac{15\epsilon_0 T}{4a}}$$

23. 00003.38

Sol. Reading = $3 + (35 \times 0.01) - (-0.03) = 3.38 \text{ mm}$

24. 00000.50

Sol. $S(t) = 5\cos(1800\pi t) + 20\cos(2000\pi t) + 5\cos(2200\pi t)$

$$= 20 \left[1 + \frac{1}{2}\cos(200\pi t) \right] \cos(2000\pi t)$$

$\therefore$ modulation index is 0.50

25. 00011.25

Sol. $\frac{1}{f} = (1.4 - 1) \left(\frac{2}{25} \right) = \frac{4}{125}$

$$f = 31.25 \text{ cm}$$

$$\text{Normal shift} = 35 - 31.25 = 3.75 \text{ cm}$$

$$\therefore 3.75 = t \left(1 - \frac{1}{\mu_2} \right) = t \left(1 - \frac{1}{1.5} \right)$$

$$\Rightarrow t = 11.25 \text{ cm}$$

26. 00006.00

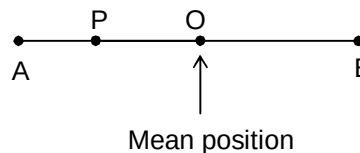
Sol. Here time period is $T = 4 \text{ sec}$

$$x = A \sin \frac{2\pi}{T} t$$

$$v = v_{\max} \cos \left(\frac{2\pi}{T} t \right)$$

Time from O to P is $t = 0.5 \text{ sec}$

$$t = \frac{T}{8}$$



$$\therefore v_{\max} = \frac{v}{\cos \frac{2\pi \cdot T}{T} \cdot \frac{T}{8}} = \frac{3\sqrt{2}}{\frac{1}{\sqrt{2}}} = 6 \text{ m/s}$$

27. 00006.00

Sol. Effective half life is

$$T_{\text{eff}} = \frac{T_1 T_2}{T_1 + T_2}$$

$$T_{\text{eff}} = \frac{(1)(2)}{1+2} = \frac{2}{3} = \frac{6}{9} \text{ hour}$$

$$\therefore k = 6$$

28. 00001.50

Sol. In container A;

$$\Delta P = \frac{n_A RT}{V} - \frac{n_A RT}{2V} = \frac{n_A RT}{2V} \quad \dots (i)$$

In container B;

$$1.5 \Delta P = \frac{n_B RT}{V} - \frac{n_B RT}{2V} = \frac{n_B RT}{2V} \quad \dots (ii)$$

By solving

$$1.5 = \frac{n_B}{n_A} = \frac{m_B}{m_A}$$

29. 00005.00

 Sol. Here $\lambda = 1\text{cm}$

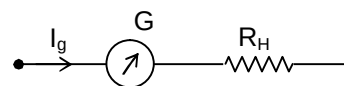
$$\therefore \frac{\lambda}{2} = 0.5\text{cm} = 5 \text{ mm}$$

30. 00200.00

 Sol. $V = I_g (R_g + R_H)$

$$2.5 = (10^{-2})(50 + R_H)$$

$$R_H = 250 - 50 = 200\Omega$$

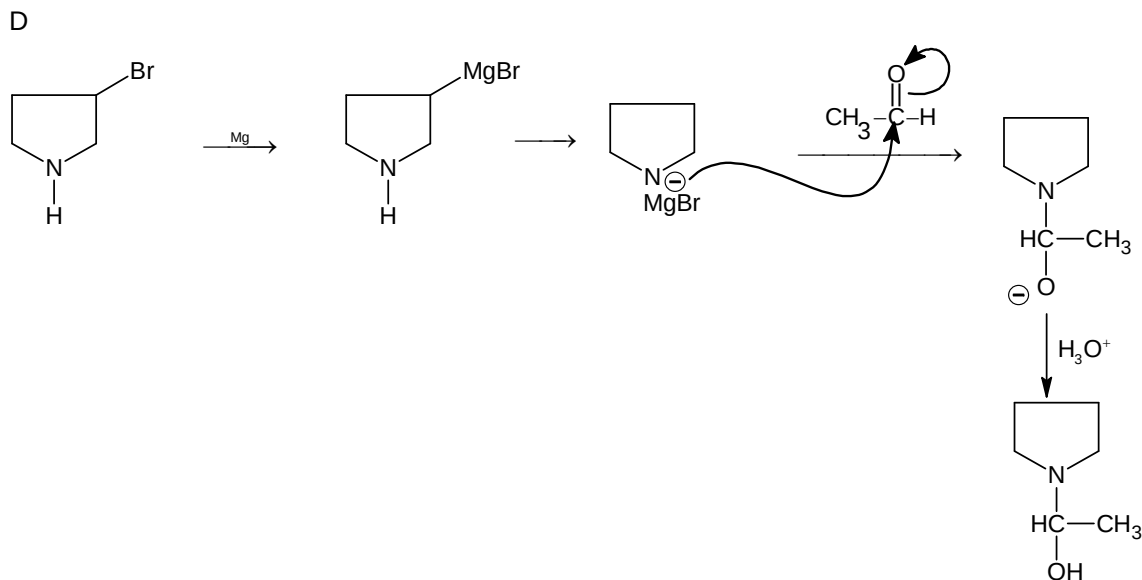


Chemistry

PART – B

SECTION – A

31.
Sol.



32.
Sol.

B
Myosin (in muscles) is fibrous protein.

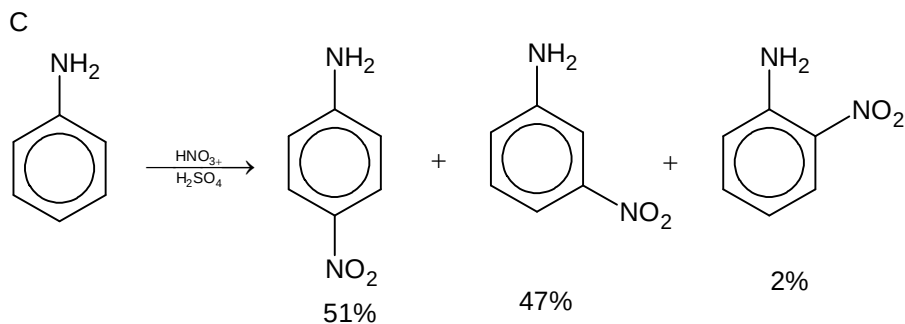
33.
Sol.

C
Oxides eg. TiO_2 sol is positively charged sol.

34.
Sol.

B
 $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ has positive value which is equal to 0.34 V.

35.
Sol.



36.
Sol.

C
Solvolysis depends on stability of carbocation.

37.
Sol.

D
 HgS , PbS and CuS are black but CdS is yellow.

38.
Sol.

A
 $\text{Co}^{2+} = 3d^6 4s^0$
 $\text{C}_2\text{O}_4^{2-}$ is chelating ligand. So, $t_{2g} = 3d^6$ and $e_g = 3d^0$

39. A

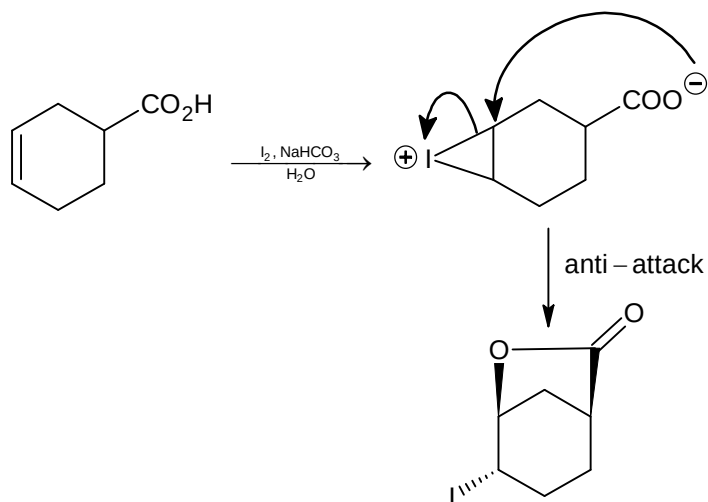
Sol. The orbitals with same number of radial nodes has same shape of curve.

40. A

 Sol. At low pressure V is high so, b can be neglected in comparison to V .

41. B

Sol.



42. D

Sol. Change in position is not allowed in resonance structures. So, 1, 2, 3 are not resonance structures, they are constitutional isomers.

43. A

 Sol. LiF is sparingly soluble due to its high lattice energy.

44. B

 Sol. $\Delta G = \Delta H + T \left(\frac{d\Delta G}{dT} \right)_P$

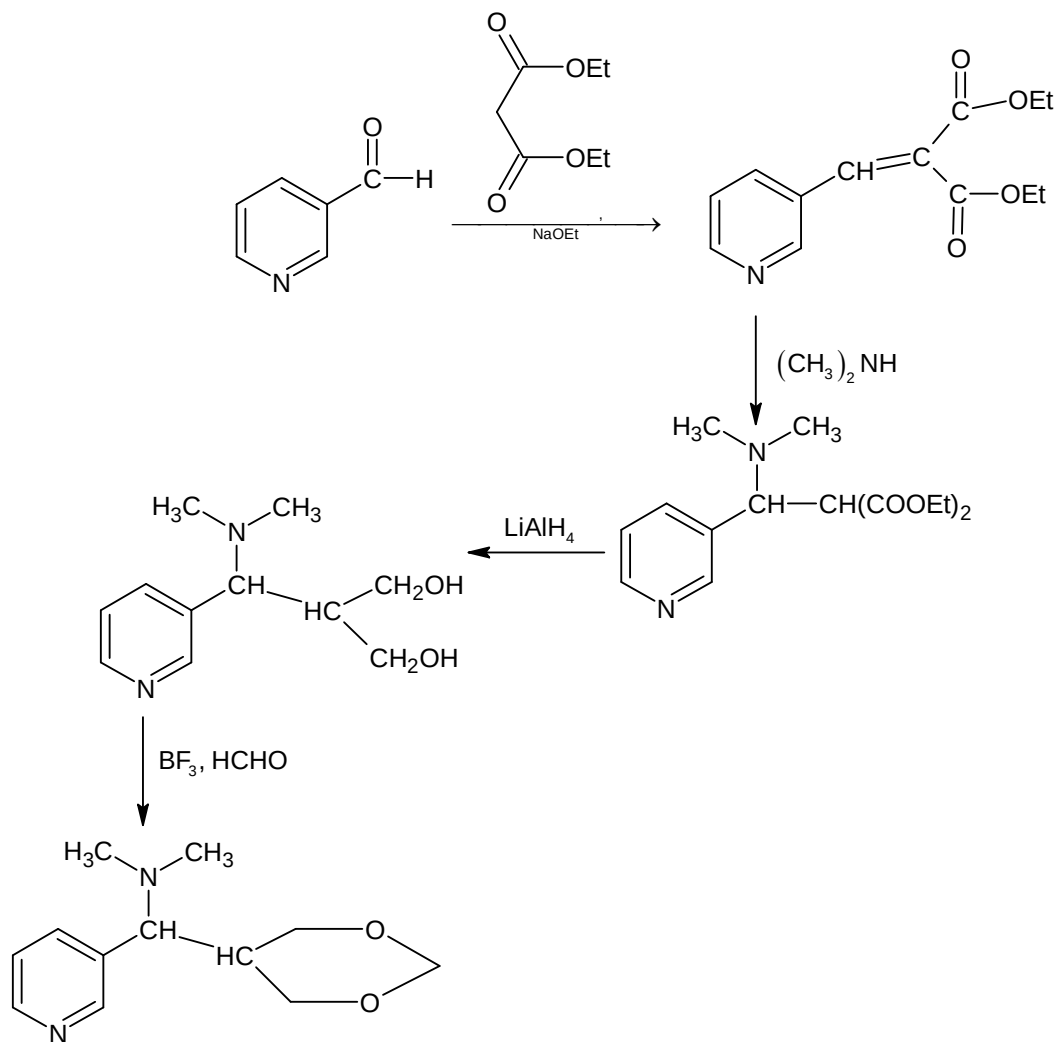
$$\Delta G = -nFE$$

$$\Delta G = \Delta H - T\Delta S$$

45. C

46. B

Sol.



47. C

Sol. CN^- shows 1,4-conjugate addition.

48. C

Sol. $\text{HCOOH} \xrightarrow{\text{conc. H}_2\text{SO}_4} \text{CO} + \text{H}_2\text{O}$
 Conc. H_2SO_4 acts as dehydrating agent.

49. B

Sol. Zone refining is useful for producing metals of very high purity, e.g. germanium, silicon, boron etc.

50. C

Sol. As size increases ionisation energy decreases.

SECTION – B

51. 00050.00

Sol. The maximum limit of nitrate in drinking water is 50 ppm.

52. 00049.26

 Sol. $\pi = iCRT$

$$\pi = 2 \times 1 \times 0.0821 \times 300$$

$$\pi = 49.26 \text{ atm}$$

53. 00001.44

 Sol. $K_a = C\alpha^2$

$$1.6 \times 10^{-5} = 0.01 \alpha^2$$

$$4 \times 10^{-2} = \alpha$$

$$\alpha = \frac{\lambda_m}{\lambda_m^0}, 4 \times 10^{-2} = \frac{\lambda_m}{360.5 \times 10^{-4}}$$

$$\lambda_m = 1442 \times 10^{-6} \text{ mho m}^2 \text{ mol}^{-1}$$

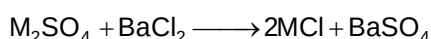
$$= 14.42 \text{ mho cm}^2 \text{ mol}^{-1}$$

$$\lambda_m = \frac{1000 \times K}{M}$$

$$K = 1.442 \times 10^{-4} \text{ mho cm}^{-1}$$

$$K = 1.442 \times 10^{-2} \text{ mho m}^{-1}$$

54. 00039.20

 Sol. $\text{MgSO}_4 + \text{BaCl}_2 \longrightarrow \text{MgCl}_2 + \text{BaSO}_4$

 Total moles of BaSO_4 = Total moles of MgSO_4 + Total moles of M_2SO_4

$$\frac{1.49}{233} = \frac{0.25}{120} + \frac{0.75}{2x + 96}$$

$$x = 39.2$$

55. 00008.00

 Sol. Number of all stereoisomers = $2^3 = 8$.

56. 00003.95

$$\text{Sol. } \rho = \frac{Z \times M}{N_A \times a^3} = \frac{4 \times 40}{6.022 \times 10^{23} \times (5.56)^3 \times (10^{-8})^3} = 1.5458$$

$$\% \text{Schottky defect} = \frac{1.5458 - 1.4848}{1.5458} \times 100 = 3.95$$

57. 00017.84

 Sol. $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$$33 = 223 - 1520 \Delta S^\circ, \quad \frac{190}{1520} = \Delta S^\circ$$

$$\Delta S^\circ = 0.125 \text{ kJ mol}^{-1} \text{ K}^{-1}$$

$$\text{For } K_{\text{eq}} = 1 \quad \Delta G^\circ = 0$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 0$$

$$\frac{\Delta H^0}{\Delta S^0} = T, \quad \frac{223}{0.125} = T$$

$$T = 1784$$

$$T = 17.84 \times 10^2$$

58. 00028.90

Sol. $R_1 = k A_1, \quad 0.04 = k A_0 e^{-k(10)}$

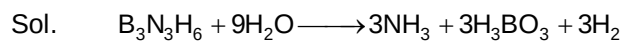
$R_2 = k A_2, \quad 0.03 = k A_0 e^{-k(22)}$

$\frac{4}{3} e^{-10k+22k}, \quad \frac{4}{3} e^{12k}$

$\ln \frac{4}{3} = 12k \quad k = \frac{1}{12} \ln \frac{4}{3}$

$t_{1/2} = \frac{0.693}{k}$

59. 00009.00



60. 00349.66

Sol. For adiabatic process $T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

$T_2 = 400K$

$W = W_1 + W_2$

$= \left(-nRT \ln \frac{V_2}{V_1} \right) + \left(\frac{nR}{\gamma-1} (T_2 - T_1) \right)$

$= 349.66 \text{ J}$

Mathematics

PART – C

SECTION – A

61. C

Sol. $L_1 = 7, L_2 = \frac{1}{2}$

62. B

Sol. $e^x \in (0, 1] \Rightarrow x \in (-\infty, 0] ; x^2 \in [0, 1] \Rightarrow x \in [-1, 1] \Rightarrow x \in [-1, 0)$
 $\Rightarrow \alpha < 0$ and $x \neq (2n+1)\pi \Rightarrow \beta = 0$

63. D

Sol. $r = 2R \Rightarrow \Delta$ is equilateral

64. A

Sol. $\sum_{i=1}^9 \alpha_i = -\frac{27}{2}$

65. C

Sol. If $(M - I)^2(M - 2I) = 0 \Rightarrow M^3 - 4M^2 + 5M - 2I = 0$
 $\Rightarrow M_1 = 2M, M_2 = 4M^3 \Rightarrow |M_1 M_2| = |8M^4| = 11^4$

66. C

Sol. Domain $\equiv (0, \infty)$, $h'(x) = \frac{1}{(1+x)} + \frac{1}{(1+x)^2} > 0 \forall x$, $h''(x) < 0 \forall x \in D$

67. C

Sol. $(x + 2022)(x + k) = -1$
 $\Rightarrow$ So, equation of hyperbola can be $x^2 - y^2 = \pm 4$

68. A

Sol. $P = M^{60}$ and $\sum a_i = M \sum \frac{1}{a_i} \Rightarrow Q = M$

69. C

Sol. $t \in (-2, 0)$

70. A

Sol. Equation of the directrix is $x - 3y + 13 = 0$

71. B

Sol. Let $a = \frac{1}{x}, b = \frac{1}{y}, c = \frac{1}{z}$
 $\Rightarrow \left(\frac{1-x}{x}\right)\left(\frac{1-y}{y}\right)\left(\frac{1-z}{z}\right) \geq 8$

72. B

Sol. ${}^9C_3 \times 2 + {}^9C_2$

73. A

$$\text{Sol. } \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = 16 \Rightarrow V = 4$$

74. D

$$\text{Sol. } (f(\theta))^2 = 16 + 25 + 2\sqrt{(16^2 + 25^2)\sin^2\theta\cos^2\theta} + 16 \times 25(\sin^4\theta + \cos^4\theta)$$

$$\max f(\theta) = \sqrt{2(16+25)}, \min f(\theta) = 4 + 5$$

75. A

$$\text{Sol. } [\vec{p} \ \vec{q} \ \vec{r}] = 4 \text{ and } |\vec{p} \cdot \vec{q}| = \frac{3}{5}$$

76. C

$$\text{Sol. } \text{Let } z_n = p_n + iq_n \Rightarrow z_{n+1} = (\sqrt{3} + i)z_n$$

77. A

$$\text{Sol. } x = 3 \text{ is the only solution}$$

78. D

$$\text{Sol. } \text{For the variate } \frac{\alpha x + \beta}{\gamma} \text{ the new standard deviation} = a \left| \frac{\alpha}{\gamma} \right|$$

79. A

$$\text{Sol. } g(x) = \tan^{-1}\left(\frac{x}{3}\right)$$

80. A

$$\text{Sol. } \text{Coefficient } x^{12} \text{ in } (x^0 + x^1 + \dots + x^9)^4$$

$$\Rightarrow {}^{15}C_3 - {}^4C_1 \times {}^5C_2 = 415$$

SECTION – B

81. 00008.00

$$\text{Sol. } \frac{m}{n} = \frac{7}{1}$$

82. 00240.00

$$\text{Sol. } \sum_{r=1}^9 r^2 - \sum_{r=1}^9 r = 285 - 45 = 240$$

83. 00100.00

$$\text{Sol. } \lambda_1 = 0, \lambda_2 = 100$$

84. 00024.50

$$\text{Sol. } z_1, z_2, \dots, z_5 \text{ will be } -1, \omega, \bar{\omega}, -\omega, -\bar{\omega}$$

$$\sum_{1 \leq i < j \leq 5} |z_i - z_j|^2 = 1 + \frac{13}{4} \times 2 + 1 + 1 + 4 + 3 + 3 + 4 + 1$$

$$\Rightarrow \lambda = 24.50$$

85. 00046.77

 Sol. Triangle formed by z_1, z_2, z_3 will be equilateral triangle

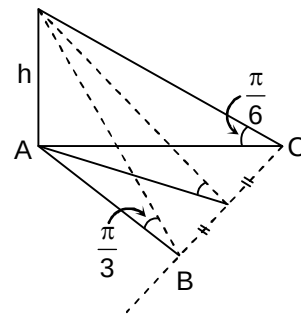
86. 00001.71

 Sol. In $\triangle AQC$

$$(QA)^2 + (QC)^2 = 2(QB^2 + BC^2)$$

$$\Rightarrow h = \frac{2}{\left(\cot^2 \frac{\pi}{3} + \cot^2 \frac{\pi}{6} - \cot^2 \frac{\pi}{4}\right)^{\frac{1}{2}}}$$

$$\Rightarrow h = \frac{12}{7}$$



87. 01024.00

 Sol. $|\text{adj } 2M| = z^6 |M|^2 = 2^{10}$

88. 00012.00

Sol. Locus will be a circle of radius two times of length of major axis

89. 00021.33

 Sol. $\frac{1}{\Delta_1} + \frac{1}{\Delta_2} = \frac{2}{\sqrt{3}a^2}$ for $y^2 = 4ax$

90. 00600.00

 Sol. Let $k = \sum_{n=1}^{20} \sum_{m=1}^{20} \tan^{-1}\left(\frac{m}{n}\right)$; $k = \sum_{n=1}^{20} \sum_{m=1}^{20} \tan^{-1}\left(\frac{n}{m}\right)$
 $\Rightarrow 2k = 400 \frac{\pi}{2} \Rightarrow k = 100\pi \Rightarrow \sum_{p=1}^3 \sum_{r=1}^p 100\pi = 600$